Gayoung Kwak

Healthcare AI / Applied Machine Learning

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PROFILE

Healthcare AI and applied machine-learning researcher combining food and nutrition domain knowledge with software and data science. Experience with clinical records, public data, and medical images, with an emphasis on reproducible evaluation and practical limitations.

EXPERIENCE

Research Intern

Cardio-Cerebrovascular Center, Severance Hospital, Yonsei University | Jul 2024 - Feb 2025

- Led an end-to-end study of LLM-based clinical feature selection for ICU delirium prediction.
- Built a leakage-aware cohort of 32,574 first ICU stays, including 2,728 incident delirium cases, and operationalized 107 features from 11 studies as 151 MIMIC-IV variables.
- Compared Scoring, Ranking, Sequencing, and log-prior procedures across GPT-4.1, Claude Sonnet 4.5, and Gemini 2.5 Pro, with five independent runs per setting.
- The best 30-feature LLM subset retained 93.0% of full-pool AUROC; supervised selectors remained stronger when outcome labels were available.

EDUCATION

B.S. Food Science and Nutrition

Kyung Hee University, Seoul Campus | 2022 - Present

B.S. Software Convergence

Kyung Hee University, Global Campus | 2024 - Present

Exchange - Data Science and Engineering

Instituto Superior Tecnico, University of Lisbon | Sep 2025 - Jan 2026

Coursework: Deep Learning, Natural Language, Information Systems and Databases, Parallel and Heterogeneous Computing Systems

RESEARCH

LLM-Based Identification of Critical Features for Delirium Prediction

Lead researcher | Manuscript under submission

Evaluated LLMs as literature-informed priors for clinical feature selection using MIMIC-IV and downstream XGBoost benchmarking.

SKILLS

Programming Python, SQL, R, C++

Cloud AWS, GCP

Web HTML/CSS, JavaScript, React

Methods Machine learning, deep learning, statistics, geospatial analysis

LANGUAGES

Korean Native

English TOEFL iBT 79 (B2)

SELECTED WORK

Show Me The Sanggwon

2026 Seoul Metropolitan Government Big Data Competition - Grand Prize

Product & Service Development, Startup Division; Mayor of Seoul Award. Led an XGBoost restaurant-site survival model using 139 features from 16 public datasets; temporal AUC 0.71, leakage controls, and SHAP explanations.

DentiScan

Data Analysis Capstone Design | 2026

Two-stage RF-DETR and Swin Transformer screening pipeline. Copy-Paste augmentation raised caries recall from 0.32 to 0.54; contributed to model experiments, backend integration, and AWS deployment.

SS-TI

Food and Nutrition Capstone Design | 2024

Built a web assessment from 40 literature-derived sugar and sodium items, evaluated with KMO and Cronbach's alpha, with personalized dietary guidance.

Book-Delivery Robot Hubs

Seoul Gwanak-gu Youth Data Science Program | 2023

Combined nine public datasets and OLS regression to prioritize library robot hubs and routes.

Food Poisoning Battle

Kyung Hee University KHUthon | 2023

Designed the concept, rules, progression, and UI for a Unity strategy game based on pathogen growth conditions.